

CYBERSECURITY PORTFOLIO

HANDS-ON LAB · CLOUD / NETWORK SECURITY

Azure Firewall Network Segmentation

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Portfolio Edition

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Other participant names removed for public portfolio use.

Network Segmentation with Azure Firewall Premium

PNC Cyber Bootcamp
Week 5 Day 3 Lab

September 2, 2026

Part 1: Establish The Baseline

Step 1: Read the Existing policy

```
PS /home/cybersec-bootcamp-2026-aug03> $RG = "rg-cybersec-cybersec-bootcamp-2026-aug03-instructor"
PS /home/cybersec-bootcamp-2026-aug03> $POLICY = "fp-cybersec-cybersec-bootcamp-2026-aug03-instructor"
```

```
PS /home/cybersec-bootcamp-2026-aug03> az network firewall policy rule-collection-group list -g $RG --policy-name $POLICY --query "[].{name:name, priority:priority}" -o table

Preview version of extension is disabled by default for extension installation, enabled for modules without stable versions.
Please run 'az config set extension.dynamic_install_allow_preview=true or false' to config it specifically.
The command requires the extension azure-firewall. Do you want to install it now? The command will continue to run after the extension is installed. (Y/n): Run 'az config set extension.use_dynamic_install=yes_without_prompt' to allow installing extensions without prompt.
This command is in preview and under development. Reference and support levels: https://aka.ms/CLI_refstatus
```

Name	Priority
rcg-default-allow	500
rcg-team1-segmentation	200

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```
PS /home/cybersec-bootcamp-2026-aug03> az network firewall policy rule-collection-group list -g $RG --policy-name $POLICY --query "[].{name:name, priority:priority}" -o table

Preview version of extension is disabled by default for extension installation, enabled for modules without stable versions.
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This command is in preview and under development. Reference and support levels: https://aka.ms/CLI_refstatus
```

Name	Priority
rcg-default-allow	500
rcg-team1-segmentation	200

```
PS /home/cybersec-bootcamp-2026-aug03> az network firewall policy rule-collection-group show -g $RG --policy-name $POLICY --name rcg-default-allow --query "ruleCollections[].{collection:name, priority:priority, action:action.type}" -o table
This command is in preview and under development. Reference and support levels: https://aka.ms/CLI_refstatus
```

Collection	Priority	Action
allow-outbound-web	600	Allow
allow-all-inter-zone	500	Allow
allow-outbound-internet	400	Allow

13.

```
[
  {
    "action": {
      "type": "Allow"
    },
    "name": "allow-outbound-web",
    "priority": 600,
    "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
    "rules": [
      {
        "description": "",
        "destinationAddresses": [],
        "fqdnTags": [],
        "httpHeadersToInsert": [],
        "name": "allow-web",
        "protocols": [
          {
            "port": 80,
            "protocolType": "Http"
          },
          {
            "port": 443,
            "protocolType": "Https"
          }
        ],
        "ruleType": "ApplicationRule",
        "sourceAddresses": [
          "10.50.1.0/24",
          "10.50.2.0/24",
          "10.50.3.0/24"
        ],
        "sourceIpGroups": [],
        "targetFqdns": [
          "*"
        ],
        "targetUrls": [],
        "terminateTLS": false,
        "webCategories": []
      }
    ]
  },
],
```

```
{
  "action": {
    "type": "Allow"
  },
  "name": "allow-all-inter-zone",
  "priority": 500,
  "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
  "rules": [
    {
      "description": "",
      "destinationAddresses": [
        "10.50.1.0/24",
        "10.50.2.0/24",
        "10.50.3.0/24"
      ],
      "destinationFqdns": [],
      "destinationIpGroups": [],
      "destinationPorts": [
        "*"
      ],
      "ipProtocols": [
        "Any"
      ],
      "name": "allow-any-any",
      "ruleType": "NetworkRule",
      "sourceAddresses": [
        "10.50.1.0/24",
        "10.50.2.0/24",
        "10.50.3.0/24"
      ],
      "sourceIpGroups": []
    }
  ]
},
{
  "action": {
    "type": "Allow"
  },
  "name": "allow-outbound-internet",
  "priority": 400,
  "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
  "rules": [
```

```
"action": {
  "type": "Allow"
},
"name": "allow-outbound-internet",
"priority": 400,
"ruleCollectionType": "FirewallPolicyFilterRuleCollection",
"rules": [
  {
    "description": "Day-0 permissive outbound. Instructor tightens during Day 1 labs.",
    "destinationAddresses": [
      "0.0.0.0/0"
    ],
    "destinationFqdns": [],
    "destinationIpGroups": [],
    "destinationPorts": [
      "*"
    ],
    "ipProtocols": [
      "TCP",
      "UDP",
      "ICMP"
    ],
    "name": "allow-any-outbound",
    "ruleType": "NetworkRule",
    "sourceAddresses": [
      "10.50.1.0/24",
      "10.50.2.0/24",
      "10.50.3.0/24"
    ],
    "sourceIpGroups": []
  }
]
}
```

Step 2: Understand what you just read

Which zones can currently reach which other zones?

- Guest: 10.50.3.0/24
- Users: 10.50.1.0/24
- Servers: 10.50.2.0/24

Which ports are permitted between them?

- destinationPorts: all ports
- ipProtocols: Any

Its description says it is a temporary Day 0 rule. What is the risk of a rule like this surviving into production and why do rules like this so often survive?

- A temporary allow all rule surviving into production creates a huge security risk because it allows different zones to communicate without proper segmentation. If a device in one zone is compromised an attacker could potentially use that access to move laterally.

Step 3: Prove it with live traffic SKIPPED

```
$  
$ ping -c 3 10.50.2.6  
PING 10.50.2.6 (10.50.2.6) 56(84) bytes of data.  
  
--- 10.50.2.6 ping statistics ---  
3 packets transmitted, 0 received, 100% packet loss, time 2085ms  
  
$ █
```

```
$ curl --max-time 10 http://10.50.2.6  
curl: (28) Connection timed out after 10003 milliseconds  
$ ping -c 3 10.50.1.5
```

```
$ ping -c 3 10.50.1.5  
PING 10.50.1.5 (10.50.1.5) 56(84) bytes of data.  
  
--- 10.50.1.5 ping statistics ---  
3 packets transmitted, 0 received, 100% packet loss, time 2040ms  
  
$ █
```

```
$ curl --max-time 10 http://10.50.2.6
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Part 2: Design the Policy on Paper

From	To	Target	Reasoning
Guest Wi-Fi	Servers	Deny	Customers must never touch banking systems
Guest Wi-Fi	Users	Deny	Guests have no business reaching staff workstations
Guest Wi-Fi	Internet	Allow	Guest Wi-Fi exists to give customers internet
Users	Servers	Allow, limited ports	Tellers need the banking app and nothing else
Servers	Users	Deny	Servers should never initiate to workstations

Part 3: Write Your Deny Rules

Step 1: Set your variables

```
VERBOSE: Authenticating to Azure ...
PS /home/cybersec-bootcamp-2026-aug03> $RG = "rg-cybersec-cybersec-bootcamp-2026-aug03-instructor"
PS /home/cybersec-bootcamp-2026-aug03> $POLICY = "fp-cybersec-cybersec-bootcamp-2026-aug03-instructor"
PS /home/cybersec-bootcamp-2026-aug03> $RCG = "rcg-team1-segmentation"
PS /home/cybersec-bootcamp-2026-aug03> $RCG = "rcg-team2-segmentation"
PS /home/cybersec-bootcamp-2026-aug03> $PRIORITY = 210
PS /home/cybersec-bootcamp-2026-aug03> █
```

Step 2: Create your rule collection group

```
PS /home/cybersec-bootcamp-2026-aug03> az network firewall policy rule-collection-group create
--resource-group $RG
This command is in preview and under development. Reference and support levels: https://aka.ms/CLI_refstatus
{
  "etag": "6cde0d40-768e-41ab-b98c-afa8ec5868a1",
  "id": "/subscriptions/307f2af0-2457-46c0-b43c-4775c45481cb/resourceGroups/rg-cybersec-cybersec-bootcamp-2026-aug03-instructor/providers/Microsoft.Network/firewallPolicies/fp-cybersec-cybersec-bootcamp-2026-aug03-instructor/ruleCollectionGroups/rcg-team2-segmentation",
  "name": "rcg-team2-segmentation",
  "priority": 210,
  "provisioningState": "Succeeded",
  "resourceGroup": "rg-cybersec-cybersec-bootcamp-2026-aug03-instructor",
  "ruleCollections": [],
  "type": "Microsoft.Network/FirewallPolicies/RuleCollectionGroups"
}
PS /home/cybersec-bootcamp-2026-aug03> █
```

Step 3: Deny Guest Wi-Fi to servers

This command is in preview and under development. Reference and s

```
{
  "ruleCollections": [
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToServers",
      "priority": 100,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.2.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-servers",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    }
  ]
}
```

PS /home/cybersec-bootcamp-2026-aug03> █

Step 4: Deny Guest Wi-Fi to server

```
{
  "ruleCollections": [
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToServers",
      "priority": 100,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.2.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-servers",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    },
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToUsers",
      "priority": 110,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.1.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-users",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    }
  ]
}
```

Step 5: Deny Servers to Users

```
{
  "ruleCollections": [
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToServers",
      "priority": 100,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.2.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-servers",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    },
    {
      "action": {
        "type": "Deny"
      },
```

```
    "type": "Deny",
    "name": "DenyGuestToUsers",
    "priority": 110,
    "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
    "rules": [
      {
        "destinationAddresses": [
          "10.50.1.0/24"
        ],
        "destinationFqdns": [],
        "destinationIpGroups": [],
        "destinationPorts": [
          "*"
        ],
        "ipProtocols": [
          "Any"
        ],
        "name": "deny-guest-users",
        "ruleType": "NetworkRule",
        "sourceAddresses": [
          "10.50.3.0/24"
        ],
        "sourceIpGroups": []
      }
    ]
  },
  {
    "action": {
      "type": "Deny"
    },
    "name": "DenyServersToUsers",
    "priority": 120,
    "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
    "rules": [
      {
        "destinationAddresses": [
          "10.50.1.0/24"
        ],
        "destinationFqdns": [],
        "destinationIpGroups": [],
        "destinationPorts": [
          "*"
        ],
        "ipProtocols": [
          "Any"
        ],
        "name": "deny-servers-users",
        "ruleType": "NetworkRule",
        "sourceAddresses": [
          "10.50.2.0/24"
        ],
        "sourceIpGroups": []
      }
    ]
  }
]
```

\$ /home/cybersec-bootcamp-2026-aug03> |

Step 6: Restrict Users to Servers to specific ports only

```
{
  "ruleCollections": [
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToServers",
      "priority": 100,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.2.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-servers",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    },
    {
      "action": {
        "type": "Deny"
      },
      "name": "DenyGuestToUsers",
      "priority": 110,
      "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
      "rules": [
        {
          "destinationAddresses": [
            "10.50.1.0/24"
          ],
          "destinationFqdns": [],
          "destinationIpGroups": [],
          "destinationPorts": [
            "*"
          ],
          "ipProtocols": [
            "Any"
          ],
          "name": "deny-guest-users",
          "ruleType": "NetworkRule",
          "sourceAddresses": [
            "10.50.3.0/24"
          ],
          "sourceIpGroups": []
        }
      ]
    }
  ]
}
```

```
,
{
  "action": {
    "type": "Deny"
  },
  "name": "DenyServersToUsers",
  "priority": 120,
  "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
  "rules": [
    {
      "destinationAddresses": [
        "10.50.1.0/24"
      ],
      "destinationFqdns": [],
      "destinationIpGroups": [],
      "destinationPorts": [
        "*"
      ],
      "ipProtocols": [
        "Any"
      ],
      "name": "deny-servers-users",
      "ruleType": "NetworkRule",
      "sourceAddresses": [
        "10.50.2.0/24"
      ],
      "sourceIpGroups": []
    }
  ]
},
{
  "action": {
    "type": "Deny"
  },
  "name": "DenyUsersToServersMgmt",
  "priority": 130,
  "ruleCollectionType": "FirewallPolicyFilterRuleCollection",
  "rules": [
    {
      "destinationAddresses": [
        "10.50.2.0/24"
      ],
      "destinationFqdns": [],
      "destinationIpGroups": [],
      "destinationPorts": [
        "22",
        "3389",
        "21",
        "23"
      ],
      "ipProtocols": [
        "TCP"
      ],
      "name": "deny-users-mgmt-ports",
      "ruleType": "NetworkRule",
      "sourceAddresses": [
        "10.50.1.0/24"
      ],
      "sourceIpGroups": []
    }
  ]
}
```

